

FROM TREND TO CYCLE: MARRIED WOMEN AND BUSINESS CYCLE RISK

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Research Question

How has the rise in married women's employment impacted U.S. business cycle dynamics and intra-household insurance?

Our Contribution

- 1 Provide a unified theory of trends and cyclical dynamics.
- 2 Show how some factors driving the trend increase cyclical volatility and others reduce it.
- 3 Quantitative evaluation of cohorts 1940-1980.

Main Findings

Theory

- Cycle: Precautionary labor supply in recessions → fewer quit out of the labor force.
 - PSID: Construct quits, not just EN.
 - SIPP: liquidity constrained households; PSID: many on the margin
- Trend: More attached, quit less → more cyclically volatile.
 - Returns to experience $\uparrow$ is key. Wage gap $\downarrow$ gives opposite.

Quantitative

- E falls more in recessions by 25% (0.8 points).
- Insurance: consumption falls 7% less if job lost in recession.
 - ... but average HH income falls 25% *more* in recessions.
- Steady state U falls by almost a full point.

Empirical Motivation and Evidence

Married Women's Employment Falls Less in Recessions

	Married women	Married men
1981	2.68%	-3.41%
1990	0.07%	-2.34%
2001	0.03%	-0.01%
2007	-2.39%	-3.46%
2020	-9.20%	-7.23%
Volatility (1976-2022)	0.09	0.45

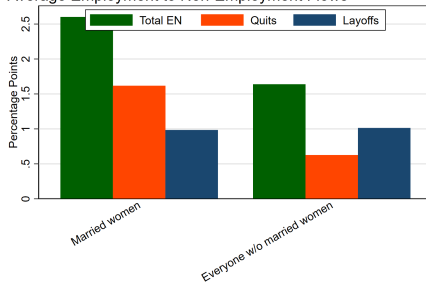
Top: Percentage point change peak to trough

Bottom: Percent deviation from trend (hours).

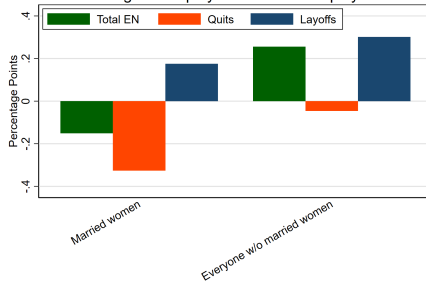
Married women have reduce aggregate hours volatility by 12% but are looking more like everyone else over time.

Quits to Nonemployment Drive Differences

Average Employment to Non-Employment Flows



Recession Change in Employment to Non-Employment Flows



Theory- Precautionary Labor Supply

Evidence non-labor income risk affects quits.

- Education as a measure of risk & asset buffer:
 - High School Men: 131% ↑ job loss in recessions
 - College Men: 87% ↑ job loss in recessions

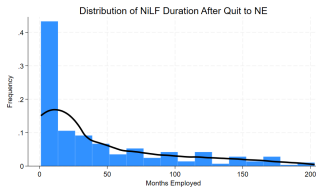
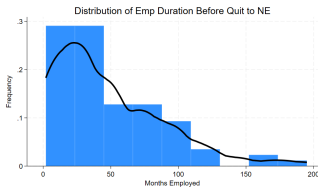
	Recession	HS Men	Col Men
1980		146%	66%
1981		88%	74%
1990		74%	57%
2001		145%	79%
2007		204%	160%

%Δ Quits: peak to trough.

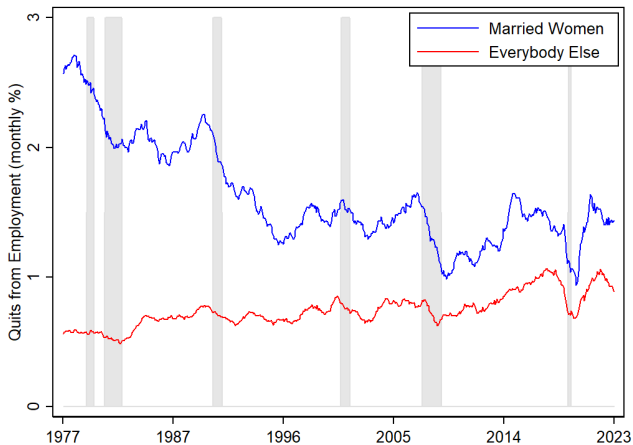
- Elasticity of married women's quit with respect to fluctuations in the unemployment rate by husband type:

High School	College
-0.1684***	0.1469
(0.0880)	(0.0902)

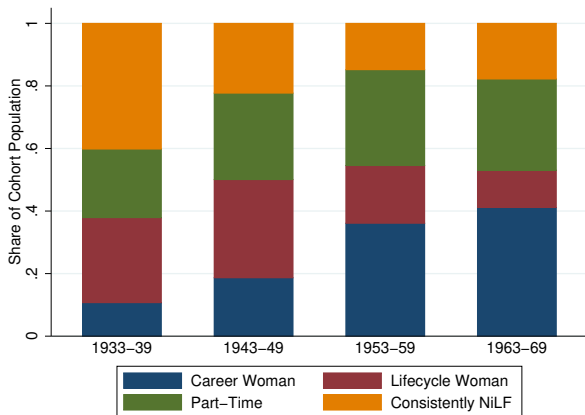
Many quits are close to the "margin".



Connecting to the Trend: Convergence in Quits



Connecting to the Trend: Higher Attachment



Careers provide a measure of distance from participation.

A Life-Cycle Model of Women's Quits & Business Cycles

- Replicates lower cyclicalities of women's employment
- Through the right mechanism: pro-cyclical quits
- With the correct cross-sectional gross flows
 - Small differences across income

Overview of Environment

- Unitary household consisting of husband and wife
- Decision problem- (extension to GE)
- Life-cycle model including child-raising years.
- Married men: newly endogenized.
 - Extension: fixed productivity and cost of work, still boring.
- Married women: make labor supply choices.
 - Employed choose hours and can quit
 - Unemployed choose search intensity
 - Vary in cost of work (ex-ante life-cycle & transitory)
- Recessions are periods of low job-finding/high job-loss.

No exogenous cyclical movement in outside option.

Model- Households' States

Households indexed by wife age $\tau \in \{0, 1, 2 \dots T\}$.

- Wife state:
 - Labor market state: $x^w \in \{E, NE\}$
 - Experience: $e \in [1, \bar{e}]$
 - Cost of work shock (transitory component): κ^T
- Husband's state:
 - Labor market state: $x^m \in \{E, U, R\}$
 - (R = recently unemployed)
- Assets: $a \in [0, \bar{a}]$
- Aggregate state: $Z \in \{0, 1\}$

Quantitative ex-ante heterogeneity

- Life-cycle fixed cost of work: $\overrightarrow{\kappa^P}$
- Permanent productivity of wife ω and husband ω^m

Model- Choices and Preferences

Choices

- Consumption (c); Savings (a)
- Time allocation of wife
 - Employed: hours h
 - Non-employed: search s
 - Residual: home production ℓ
- Whether the wife quits

Preferences

$$U(c, h) - \mathbf{1}_{s=E} \kappa$$

- $\kappa = \kappa^P + \kappa^T$, wife's total fixed cost of work.

Model-Resources

$$c + a' \leq w(\omega, e; Z)h + f(\omega)\ell^\nu + y^m(s^m; Z) + Ra$$

Market Income

- Wife: $w(\omega, e; Z)h$
 - Wage a function of productivity, experience, and the cycle
- Husband: $y^m(\tau, \omega^m, x^m; Z)$
 - Income a function of wife's age, productivity, labor market state, and the cycle

Home Production: $f(\omega)\ell^\nu$

- Home productivity related to market via fixed type ω .
- **Key: many women across the income spectrum on the margin.**

Asset Income: exogenous interest R

Model- Transitions

Labor Market

- Wife
 - Job Loss (exogenous): $\lambda^u(Z)$
 - Job Finding (endogenous search efficiency): $s^\eta \lambda^f(Z)$
- Husband- Markov $\pi^m(x'^m|x^m; Z)$

Wife's Experience

- Weakly increasing in hours during employment: $e' = g(e, h)$
- Weakly decreasing during unemployment: $e' = g^N(e)$

Ageing τ is stochastic

Aggregate Z is Markov

What Drives Labor Supply?

Less likely to exit when

- Marginal utility of consumption rises (wealth effect)
 - Husband or asset income is low
- Fixed cost of work is low
- Wage at work relative to home productivity is high
- Dynamic gain is high-
 - Returns to experience are high
 - Precautionary: expect MU to be high in future

What Drives Labor Supply?

A woman indifferent between continuing working and exiting

$$u(c^E, h) - \kappa + \beta EV(g(e, h), a'^E, x'^m; Z') = u(c^N) + \beta EV^N(g^N(e), a'^N, x'^m; Z')$$

$$c^E + a'^E \leq w(e, \omega; Z)h + f(\omega)(1 - h)^\nu + y^m(x^m, Z) + Ra$$

$$c^N + a'^N \leq f(\omega) + y^m(x^m, Z) + Ra$$

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Cross-Section: women of all income levels work

Life-cycle: women of all income types exit for child care

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Recession-

- Precautionary Labor Supply: higher likelihood y^H falls
- Job Hoarding: lower job finding rate
- Lower Wages: works in opposite direction

What Drives Labor Supply? Summary.

Pro-cyclical Quits

- Insurance: both realized risk and precautionary.
- Job hording
- Wages (work in the opposite direction)

Trend to cycle- wage gap, returns to experience, cost of work

- All change who is the marginal participant, and how many
- Closing wage gap reduces effect of husband's income risk
- Increasing returns to experience reduces the importance of transitory factors to participation
- Lower fixed cost of work allows to change hours without quitting

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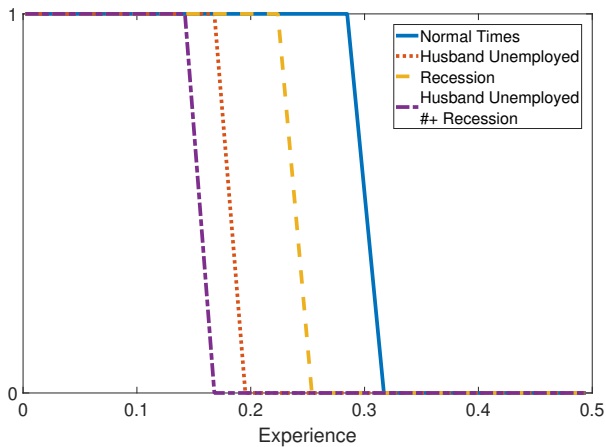
Hours and search not (as) counter-cyclical

» Hours

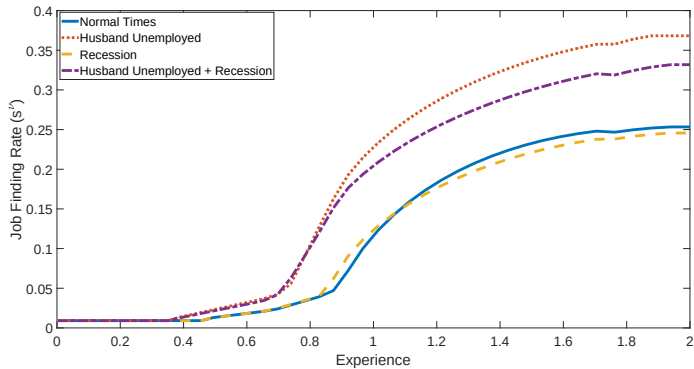
» Search

- Opportunity cost is home production, not leisure
- Hours - fall in wages counters insurance motives
- Entry small/acyclical
 - Jobs are hard to find in a recession

Procyclical Quits



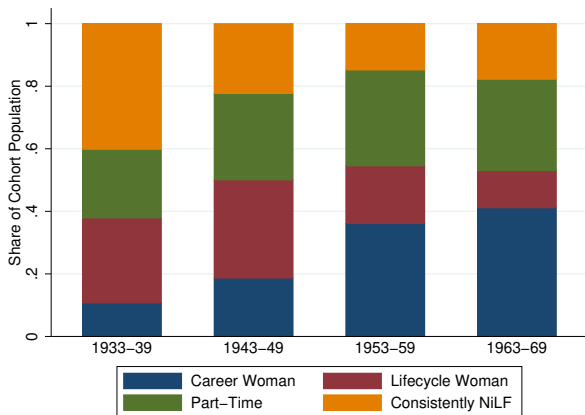
Acyclical Entry.



Calibration

Careers provide a distribution of labor supply elasticities.

- **Career Women:** annual hours $\geq 1,500$
- **Part-time:** $400 \geq$ annual hours $< 1,500$
- **NiLF:** annual hours < 400
- **Lifecycle:** switch from NiLF to Career



Calibration

$$u(c, h; \kappa) = \frac{c^{1-\gamma}}{1-\gamma} - \mu_h \frac{h^{1+\eta}}{1+\eta} - \kappa$$

- Impact of non-labor income on labor supply
- Fixed cost- permanent $\ln(\bar{\kappa}) \sim \mathcal{N}$; lifecycle $\kappa_m \sim U[1, \bar{\kappa}_m]$
- Wages $w(\omega, e; Z) = \phi(Z)\tau_w\omega(1 + \gamma_e e^{\xi_e})$
 - Cohort-specific wage gap within couples- τ_w
 - Cohort-specific returns to experience- γ_e
 - Business cycles- $\phi(Z)$
- Home Production: $f(\omega) = \bar{y}_h + z_h(\omega)^{\alpha_h}$

Calibration

Most important part: get many women on the margin!

- Historic gross participation exits are large: 5% per month.
- Life-cycle is a key factor, but not enough.
- Flows from across the income distribution
 - Low cross-sectional wage elasticity
 - Need to amp up effects of husband's

Our solution

- Low own wage elasticity
 - SE would be large since IE is small due to husband's wage
 - Home production tech increases opportunity cost with labor market productivity, thus lowering SE.
 - Cross-Section: Home tech increasing in fixed labor productivity
 - Panel: Concavity of home tech lowers response of hours/search
- High elasticity to husband's income
 - Non participants have high potential wage.
 - Fixed cost wedge. (To be fair, this is latent.)
- Low Frisch
 - Preferences can be balanced growth. (IE and SE cancel)

Aside: Precautionary Labor Supply or Savings?

Precautionary labor supply operates like precautionary savings

- Identical theory of prudence wrt non-labor income risk.
- Subst. effect is an amplifier wrt to wage risk.

Quantitative interactions

- Empirical evidence favors insurance from labor not savings.
 - Blundell, Pistaferri, & Saporta-Eksten (2016)- 81% of insurance available to husband's wage shocks is by family labor supply.
 - Liquidity constraints
- Precautionary labor supply feeds buffer stock savings.
 - We choose $R = 0$ to lower savings.

Calibration- Wages & Experience

$$w(\omega, e; Z) = \phi(Z)\tau_w\omega(1 + \gamma_e e^{\xi_e})$$

- PSID with Heckman correction on ordered probit
 - Fulltime, Part-time, non employed
 - Exclusion- demographics, children, lags of husband's income
- $\phi(\text{recession}) = 0.88$
- $\xi_e = 0.8$
- $\gamma_e = 0.5$

$$e' = (1 - \delta_e)e + \theta_e e h^{\psi_e}$$

- $\delta_e = 0.005$
- $\theta_e = 0.025$
- $\psi_e = 0.66$

Fixed component ω & cohort gap τ_w estimated w/in model

Calibration-Time Allocation/Participation

- Labor: $u(c, h; \kappa) = \frac{c^{1-\gamma}}{1-\gamma} - \mu_h \frac{h^{1+\eta}}{1+\eta} - \kappa$
- Search Efficiency: $s^\nu \lambda^f(Z)$
- Home production: $f(w) \ell^{\nu_h}$

Estimates

- $\nu = 0.5$, set in business cycle lit
- $\eta = 1.4$
- $\nu_h = 0.65$
- $\bar{\kappa} \in [0, 0.075]$; $\kappa^m \in [1, 2.27]$

Implications

- Job finding efficiency is 15% lower in recession
- Exog component of job loss is 1.7x higher in recession
- Average intensive cost $\sim 6\%$ consumption tax
- Average extensive cost $\sim 10.2\text{-}16.4\%$ consumption tax

Labor Supply Targets & Fit

Moment	Data	Model
Wage Gap	0.66	0.66
St Dev Fixed Effect in $\ln(\text{Wages})$	0.37	0.37
av hours of employed	40%	40%
<u>Career Shares</u>		
Lifecycle	31%	33%
Part Time	28%	24%
Career	19%	17%
Mostly NILF	22%	25%
<u>Cyclical Participation Flows</u>		
tot E-NonE expansion	5.0%	4.6%
tot E-NonE recession	4.8%	4.4%
quit E-NonE expansion	3.4%	3.5%
quit E-NonE recession	2.8%	2.6%
Δ Epop recession	-1.7%	-1.8%

Contribution to Cyclicalities by Career Type

	Change in Recession		Share of Pop
	Hours	Employment	
Career	-4.3%	-3.3%	17%
Part-Time	-3.5%	-2.8%	33%
Mostly NiLF	-1.2%	-0.8%	21%
Life-Cycle	-0.9%	-0.8%	29%
All	-2.0%	-1.8%	100%

1940s cohort

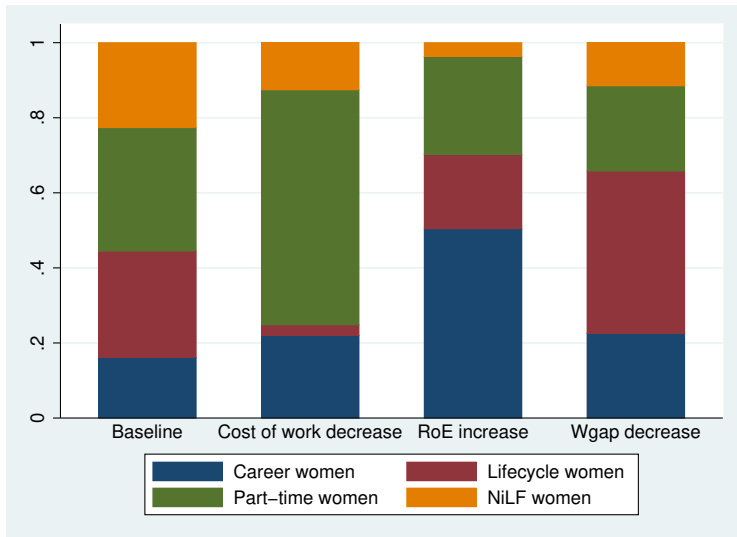
From Trend to Cycle

Calibrate- to 1940's cohort

Experiments- Generate increase in employment and check implications

- ① Increase returns to tenure (observed)
 - ② Decrease gender wage gap (observed)
 - ③ Decrease fixed cost of work (residual)
- First: all change in employment from one factor
 - Next: cohort accounting

From Trend to Careers



From Trend to Cycle

Percentage point change *relative to the 1940s baseline*

- Drop in employment during recessions
- Share of the population

	Return to Exp ↑		Wage Gap ↓		Cost of Work ↓	
	Empl	Pop Share	Empl	Pop Share	Empl	Pop Share
Part-Time	+0.5	-5	-0.4	-5	-0.1	+40
Career	+0.2	+33	-0.2	+4	+0.3	+4
Life-Cycle	+0.8	-10	-0.1	+12	-0.7	-27
Mostly NiLF	+0.4	-18	0.0	-11	+0.4	-17
All	+0.7		-1.4		+1.8	

Cohorts

	Clinton 1940	Winfrey 1950	Obama 1960	Milano 1970	Rapinoe 1980
Wage Gap	0.71	0.74	0.77	0.76	0.77
Return to Expr (γ)	0.50	0.55	0.58	0.68	0.69
Empl Rate	0.62	0.67	0.71	0.73	0.72
Cost vs 1940s	1.0	1.01	0.88	1.0	1.04

Table: Parameters and Targets

Cohorts

	Clinton 1940		Winfrey 1950		Obama 1960		Milano 1970		Rapinoe 1980	
	Model	Data	Model	Data	Model	Data	Model	Data	Model	Data
Part-Time	0.24	0.28	0.23	0.30	0.26	0.30	0.12	-	0.18	-
Career	0.17	0.19	0.27	0.28	0.40	0.40	0.38	-	0.37	-
Life-Cycle	0.33	0.31	0.31	0.18	0.14	0.12	0.30	-	0.27	-
Mostly NiLF	0.25	0.22	0.19	0.17	0.19	0.19	0.20	-	0.19	-
U-rate	7.2		6.1		4.9		4.7		4.9	

Table: Untargetted moments.

Cohorts

Cohort	Employment/Pop Δ Recess		Quit E to Non-E Δ Recess		Wife Share of Income Δ Recess	
1940	62%	-1.7%	3.5%	-33.0%	22.5%	+2.5%
1950	67	-5.9	2.5	-30.0	26.2	+2.3
1960	71	-4.6	1.8	-25.5	29.9	+2.3
1970	73	-7.2	2.3	-26.7	27.6	+2.1
1980	72	-6.3	2.4	-29.6	27.2	+2.3

Table: Cohort predictions for the business cycle.

Cohorts

Cohort	Change in Household Income Relative to Fall for 1940's cohort	
	Δ Recession	Δ Husband's Job loss
1940 (normalized)	-100%	-100%
1950	-115.2%	-94.9%
1960	-116.2%	-92.5%
1970	-124.2%	-93.9%
1980	-125.6%	-94.6%
1980's decomposed		
Wage gap only	-92.1%	-97.6%
Return to Exp. only	+114.8%	-87.1%
Fixed Cost only	+119.2%	-91.2%

Table: Changes in income volatility relative to the 1940s cohort.

Summary + Future work

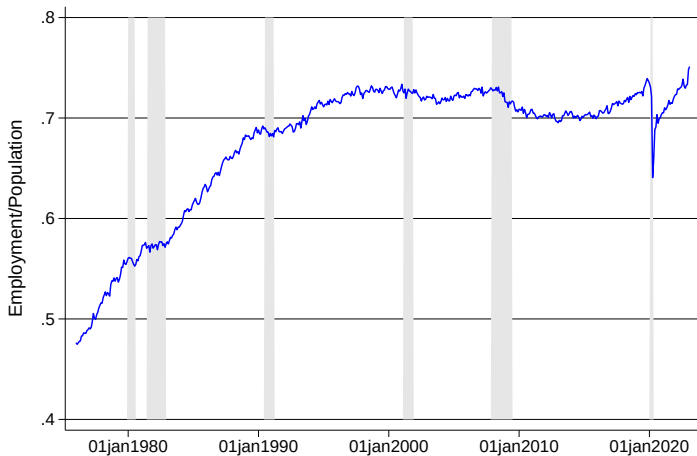
How has the rise in married women's employment impacted U.S. business cycles and intra-household insurance?

- Life-cycle model of women's labor supply and business cycles.
- Three empirically evident theories of participation trends.
- Each has unique implications for
 - Change in married women's career paths
 - Change in business cycle volatility

Big picture:

- Attachment driven employment growth = higher volatility
- Quantitatively relevant to labor market trends:
 - Brings down U by almost a full point
 - E falls more in recessions by 25% (0.8 points).

Married Women's Employment



Married Women's Cyclical Volatility of Employment

	Total	Married		Single	
		Women	Men	Women	Men
1976-2022					
Cyclical volatility	0.1334	0.0561	0.1579	0.1420	0.2518

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Employment Declines in Recessions

	Married women	Married men	Single women	Single men
1981	2.68%	-3.41%	-0.09%	-4.18%
1990	0.07%	-2.34%	-0.05%	-3.70%
2001	0.03%	-0.01%	-1.32%	-1.11%
2007	-2.39%	-3.46%	-3.92%	-6.39%
2020	-9.20%	-7.23%	-11.74%	-11.17%

Percentage point changes for employment-population ratio from peak to trough

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What Drives Labor Supply?

$$\begin{array}{c}
 \text{utility cost} \qquad \qquad \qquad \text{current gain} \\
 \underbrace{v'(h)} = \underbrace{u'(w(e, Z)h + f(\omega, (1 - h)) + y^m(s^m; Z))[w(e, \omega; Z) + f_h(\omega, (1 - h))]}_{\text{future gain}} \\
 \quad + \underbrace{\beta E[V_e(g(e, h), a', x'^m, Z'))g_h(e, h)]}_{\text{future gain}}
 \end{array}$$

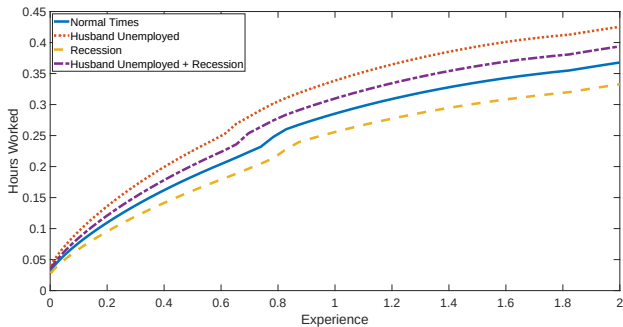
Work more when:

- Current marginal utility (MU) is high:
 - Husband or asset income is low
- Wage at work relative to home productivity is high
- Dynamic gain is high-
 - Returns to experience are high
 - Precautionary: expect MU to be high in future

What Drives Labor Supply?

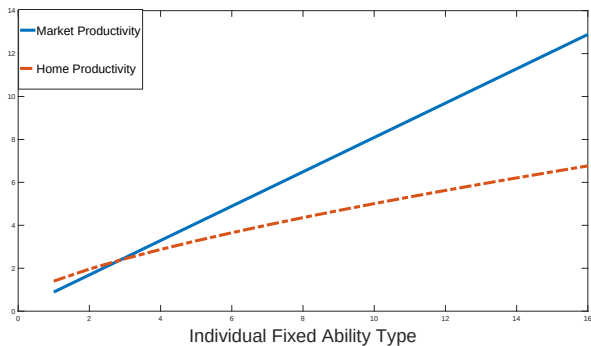
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Hours Choice



» Back

Home vs Market Tech



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Model- Full Programming Problems

Employed wife- intensive & quit choice

$$\begin{aligned} V_{\tau}^E(e, a; x^m, Z) &= \max_{h, c, a'} U(c, h) - \kappa \\ &\quad + \beta(1 - \lambda^u(Z))E_{\tau', x'^m, Z'} \max\{V_{\tau'}^E(e', a'; x'^m, Z'), V_{\tau'}^N(e', a'; x'^m, Z')\} \\ &\quad + \beta\lambda^u(Z)E_{\tau', x'^m, Z'} V_{\tau'}^N(e', a'; x'^m, Z') \\ \text{st } c + a' &\leq w(e, \omega; Z)h + f(\omega)(1 - h)^{\nu} + y^m(x^m; Z) + Ra \\ e' &= g(e, h) \quad a' \geq 0 \quad h \in (0, 1] \end{aligned}$$

Non-employed wife- extensive (search)

$$\begin{aligned} V_{\tau}^N(e, a; x^m, Z) &= \max_{s, c, a'} u(c) \\ &\quad + \beta E_{\tau', x'^m, Z'} [(1 - s\lambda^f(Z))V_{\tau'}^N(e', a'; x'^m, Z') + s\lambda^f(Z)V_{\tau'}^E(e', a'; x'^m, Z')] \\ \text{st } c + a' &\leq f(\omega)(1 - s)^{\nu} + y^m(x^m, Z) + Ra \\ e' &= g^N(e) \quad a' \geq 0 \quad s \in [0, 1] \end{aligned}$$